

Natural Construction of Homography Transformation

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TOC

- Description
- Nomenclature
- Problem
- Existing Solutions
 - $\|h\|=1$
 - $h_{33} = 1$
 - Paul Heckbert
- Elements
 - Translation
 - Linear
 - Simple Homography
- Building
 - Elements
 - Assembling
- Assembling
- Higher Dimensions
- Constraints
- Shortcut
- Miscellaneous
 - Source code
 - Off topic
 - Links

Description

Computing the plane to plane homography without fancy thing such as inversion of big matrix or eigenvalues.
In here it will be present method in step-by-step manner where every step has some geometric meaning.

Nomenclature

2D Euclid vector

Ordered pair of two real numbers.

$$\mathbf{p} = (x, y) = [x, y]^T; x, y \in \mathbb{R}$$

2D Homography point

Ordered triplet of three real numbers.

$$\mathbf{p} = (x, y, z) = [x, y, z]^T; x, y, z \in \mathbb{R}$$

Homography

$$\mathbf{H}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

Definition: project (first) plane to (second) plane.

Matrix form

Algebraic form

$$\left[\begin{array}{cc|c} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ \hline h_{31} & h_{32} & h_{33} \end{array} \right] \quad \begin{aligned} x' &= \frac{x \cdot h_{11} + y \cdot h_{12} + h_{13}}{x \cdot h_{31} + y \cdot h_{32} + h_{33}} \\ y' &= \frac{x \cdot h_{21} + y \cdot h_{22} + h_{23}}{x \cdot h_{31} + y \cdot h_{32} + h_{33}} \end{aligned}$$

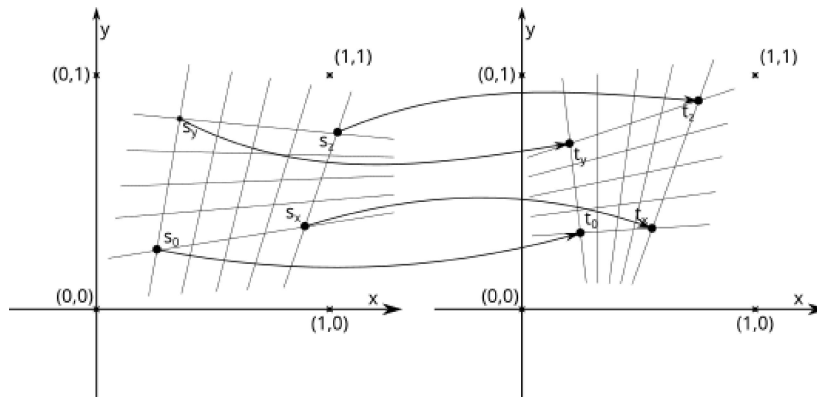
Problem

What we have:

- $s_0, s_x, s_y, s_z \in \mathbb{R}^2$
- $t_0, t_x, t_y, t_z \in \mathbb{R}^2$
- $o=(0,0)$
- $area(s_i, s_j, s_k) \neq 0; i \neq j \neq k; i, j, k \in \{0, x, y, z\}$
- $area(t_i, t_j, t_k) \neq 0; i \neq j \neq k; i, j, k \in \{0, x, y, z\}$
- Nice to have:
 $|area(t_i, t_j, t_k)| \leq area(t_0, t_x, t_y); i, j, k \in \{0, x, y, z\}$
 $|area(s_i, s_j, s_k)| \leq area(s_0, s_x, s_y); i, j, k \in \{0, x, y, z\}$

What we want:

- $\mathbf{H}$ is homography.
- $t_i = \mathbf{H}(s_i); i \in \{0, x, y, z\}$



Existing Solutions

$$\|h\|=1$$

For simplicity in here $x_1 = s_{0_x}, y_1 = s_{0_y}, \dots, x'_1 = t_{0_x}, y'_1 = t_{0_y}, \dots$

$$\begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 & -x_1 x'_1 & -y_1 x'_1 & -x'_1 \\ 0 & 0 & 0 & x_1 & y_1 & 1 & -x_1 y'_1 & -y_1 y'_1 & -y'_1 \\ x_2 & y_2 & 1 & 0 & 0 & 0 & -x_2 x'_2 & -y_2 x'_2 & -x'_2 \\ 0 & 0 & 0 & x_2 & y_2 & 1 & -x_2 y'_2 & -y_2 y'_2 & -y'_2 \\ x_3 & y_3 & 1 & 0 & 0 & 0 & -x_3 x'_3 & -y_3 x'_3 & -x'_3 \\ 0 & 0 & 0 & x_3 & y_3 & 1 & -x_3 y'_3 & -y_3 y'_3 & -y'_3 \\ x_4 & y_4 & 1 & 0 & 0 & 0 & -x_4 x'_4 & -y_4 x'_4 & -x'_4 \\ 0 & 0 & 0 & x_4 & y_4 & 1 & -x_4 y'_4 & -y_4 y'_4 & -y'_4 \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \\ h_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Ah=0 \text{ and } \|h\|=h_{11}^2 + h_{12}^2 + \dots + h_{33}^2 = 1$$

...

$$A^T A h = \lambda h$$

Eigenvector h with smallest eigenvalue λ of matrix $A^T A$.

$$h_{33} = 1$$

For simplicity in here $x_1 = s_{0_x}, y_1 = s_{0_y}, \dots, x'_1 = t_{0_x}, y'_1 = t_{0_y}, \dots$

And we have to solve :

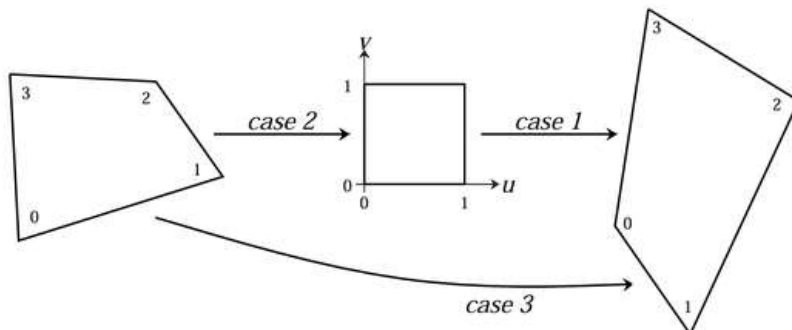
$$\begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 & -x_1 x'_1 & -y_1 x'_1 \\ 0 & 0 & 0 & x_1 & y_1 & 1 & -x_1 y'_1 & -y_1 y'_1 \\ x_2 & y_2 & 1 & 0 & 0 & 0 & -x_2 x'_2 & -y_2 x'_2 \\ 0 & 0 & 0 & x_2 & y_2 & 1 & -x_2 y'_2 & -y_2 y'_2 \\ x_3 & y_3 & 1 & 0 & 0 & 0 & -x_3 x'_3 & -y_3 x'_3 \\ 0 & 0 & 0 & x_3 & y_3 & 1 & -x_3 y'_3 & -y_3 y'_3 \\ x_4 & y_4 & 1 & 0 & 0 & 0 & -x_4 x'_4 & -y_4 x'_4 \\ 0 & 0 & 0 & x_4 & y_4 & 1 & -x_4 y'_4 & -y_4 y'_4 \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \\ x_2 \\ y_2 \\ x_3 \\ y_3 \\ y_4 \\ x_4 \end{bmatrix}$$

Main disadvantaged here is that h_{33} can be near zero and possibly produce numerical instabilities.

Paul Heckbert

Same idea, similar principle.

- Assume that $h_{33} = 1$.
- Direct calculation from quadrilateral to square and vice versa.
- Fastest closed form solution.



- This image was taken from [Heckbert99]. See also [Heckbert89].

Elements

... or what we need to make matrix.

Translation

Move for some vector.

$$(x+X, y+Y) = \mathbf{P}(x, y)$$

Homography matrix:

$$\left[\begin{array}{cc|c} 1 & 0 & X \\ 0 & 1 & Y \\ \hline 0 & 0 & 1 \end{array} \right]$$

Linear

If $\mathbf{L}$ is linear function then:

$$\mathbf{L}(\alpha \cdot \mathbf{x} + \beta \cdot \mathbf{y}) = \alpha \cdot \mathbf{L}(\mathbf{x}) + \beta \cdot \mathbf{L}(\mathbf{y})$$

Represented using homography matrix (with one of many possible decomposition):

$$\left[\begin{array}{cc|c} a_{11} & a_{12} & 0 \\ a_{12} & a_{22} & 0 \\ \hline 0 & 0 & 1 \end{array} \right] = \left[\begin{array}{cc|c} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ \hline 0 & 0 & 1 \end{array} \right] \left[\begin{array}{cc|c} s_x & 0 & 0 \\ 0 & s_y & 0 \\ \hline 0 & 0 & 1 \end{array} \right] \left[\begin{array}{cc|c} 1 & \tan(\beta) & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right]$$

$\alpha, s_x, s_y, \beta \in \mathbb{R}$

Simple Homography

$$(0,0) = \mathbf{P}(0,0);$$

$$(1,0) = \mathbf{P}(1,0);$$

$$(0,1) = \mathbf{P}(0,1);$$

$$(X,Y) = \mathbf{P}(1,1);$$

Homography matrix:

$$\left[\begin{array}{cc|c} X & 0 & 0 \\ 0 & Y & 0 \\ \hline 1-Y & 1-X & X+Y-1 \end{array} \right]$$

$$0 \neq (1-Y)^2 + (1-X)^2 + (X+Y-1)^2$$

Always $\neq 0$.

Always exists.

If we observe as function we have minimum at $(X,Y)=(2/3,2/3)$.

... and vice versa

$$(0,0) = \mathbf{P}(0,0);$$

$$(1,0) = \mathbf{P}(1,0);$$

$$(0,1) = \mathbf{P}(0,1);$$

$$(1,1) = \mathbf{P}(X,Y);$$

Homography matrix:

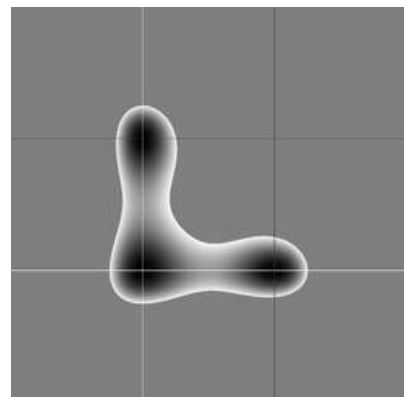
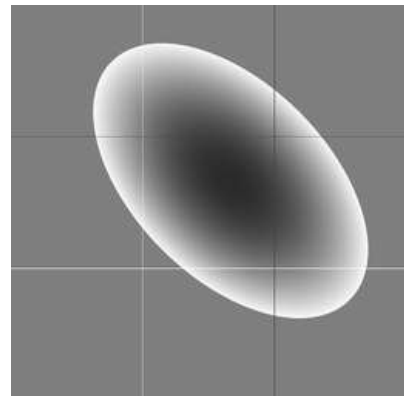
$$\left[\begin{array}{cc|c} Y \cdot (X+Y-1) & 0 & 0 \\ 0 & X \cdot (X+Y-1) & 0 \\ \hline Y \cdot (Y-1) & X \cdot (X-1) & X \cdot Y \end{array} \right]$$

$$0 \neq (Y \cdot (Y-1))^2 + (X \cdot (X-1))^2 + (X \cdot Y)^2$$

$$(X,Y) \neq (0,0)$$

$$(X,Y) \neq (1,0)$$

$$(X,Y) \neq (0,1)$$



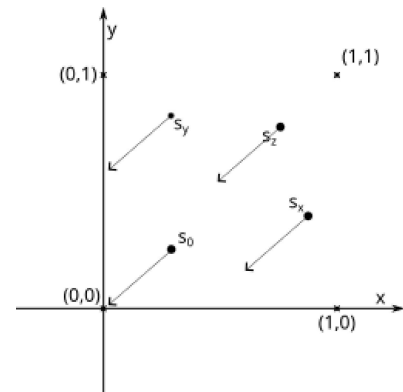
Building

Elements

$\mathbf{T}_s$: translation matrix, translate from s_0 to $o=(0,0)$

$$\mathbf{T}_s = \left[\begin{array}{cc|c} 1 & 0 & -s_{0x} \\ 0 & 1 & -s_{0y} \\ \hline 0 & 0 & 1 \end{array} \right]$$

-()	+	-	·
2	0	0	0



$\mathbf{L}_s$:

$$(0,0) = \mathbf{L}_s((0,0)),$$

$$(1,0) = \mathbf{L}_s(\mathbf{T}_s(s_x)),$$

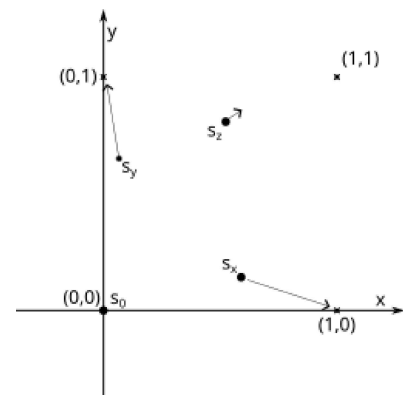
$$(0,1) = \mathbf{L}_s(\mathbf{T}_s(s_y))$$

$$a_x = \mathbf{T}_s(s_x)$$

$$a_y = \mathbf{T}_s(s_y)$$

$$\mathbf{L}_s = \left[\begin{array}{cc|c} a_{yy} & -a_{yx} & 0 \\ -a_{xy} & a_{xx} & 0 \\ \hline 0 & 0 & a_{xx} \cdot a_{yy} - a_{xy} \cdot a_{yx} \end{array} \right]$$

-()	+	-	·
0+2	0+0	4+1	0+2



$\mathbf{P}_s$:

$$(0,0) = \mathbf{P}_s((0,0)),$$

$$(1,0) = \mathbf{P}_s((1,0)),$$

$$(0,1) = \mathbf{P}_s((0,1)),$$

$$(1,1) = \mathbf{P}_s(\mathbf{L}_s \mathbf{T}_s(s_z))$$

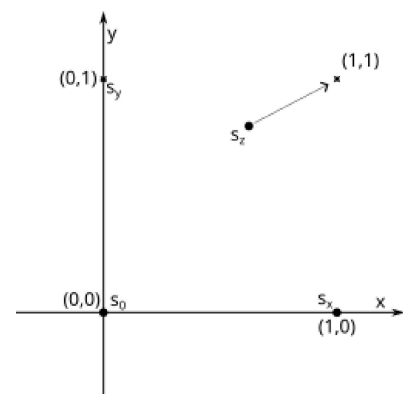
$$a_z = \mathbf{L}_s \mathbf{T}_s(s_z)$$

$$X = a_{z_x}$$

$$Y = a_{z_y}$$

$$\mathbf{P}_s = \left[\begin{array}{cc|c} Y \cdot (X + Y - 1) & 0 & 0 \\ 0 & X \cdot (X + Y - 1) & 0 \\ \hline Y \cdot (Y - 1) & X \cdot (X - 1) & X \cdot Y \end{array} \right]$$

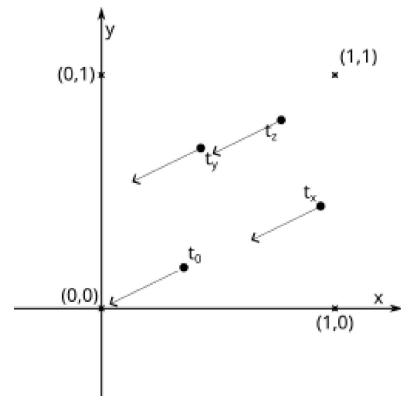
-()	+	-	·
0+0+0	0+2+2	2+0+4	0+4+4



$\mathbf{T}_t$: translation matrix, translate from t_0 to o

$$\mathbf{T}_t = \left[\begin{array}{cc|c} 1 & 0 & -t_{0x} \\ 0 & 1 & -t_{0y} \\ \hline 0 & 0 & 1 \end{array} \right] \quad \mathbf{T}_t^{-1} = \left[\begin{array}{cc|c} 1 & 0 & +t_{0x} \\ 0 & 1 & +t_{0y} \\ \hline 0 & 0 & 1 \end{array} \right]$$

$$\mathbf{T}_t^{-1} : \begin{array}{|c|c|c|c|} \hline -() & + & - & \cdot \\ \hline 0 & 0 & 0 & 0 \\ \hline \end{array}$$



$\mathbf{L}_t$:

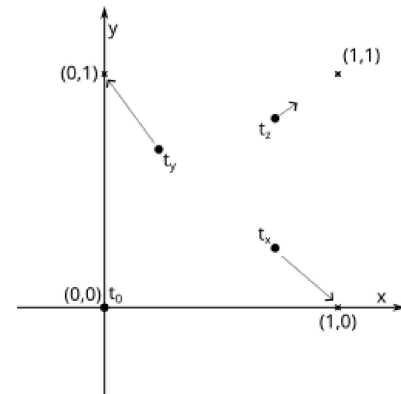
$$\begin{aligned} (0,0) &= \mathbf{L}_t((0,0)), \\ (1,0) &= \mathbf{L}_t(\mathbf{T}_t(t_x)), \\ (0,1) &= \mathbf{L}_t(\mathbf{T}_t(t_y)) \end{aligned}$$

$$b_x = \mathbf{T}_t(t_x)$$

$$b_y = \mathbf{T}_t(t_y)$$

$$\mathbf{L}_t = \left[\begin{array}{cc|c} b_{yy} & -b_{yx} & 0 \\ -b_{xy} & b_{xx} & 0 \\ \hline 0 & 0 & b_{xx} \cdot b_{yy} - b_{xy} \cdot b_{yx} \end{array} \right] \quad \mathbf{L}_t^{-1} = \left[\begin{array}{cc|c} b_{xx} & b_{yx} & 0 \\ b_{xy} & b_{yy} & 0 \\ \hline 0 & 0 & 1 \end{array} \right]$$

$$\mathbf{L}_t^{-1} : \begin{array}{|c|c|c|c|} \hline -() & + & - & \cdot \\ \hline 0+0 & 0+0 & 4+0 & 0 \\ \hline \end{array}$$



$\mathbf{P}_t$:

$$\begin{aligned} (0,0) &= \mathbf{P}_t((0,0)), \\ (1,0) &= \mathbf{P}_t((1,0)), \\ (0,1) &= \mathbf{P}_t((0,1)), \\ (1,1) &= \mathbf{P}_t(\mathbf{L}_t \mathbf{T}_t(t_z)) \end{aligned}$$

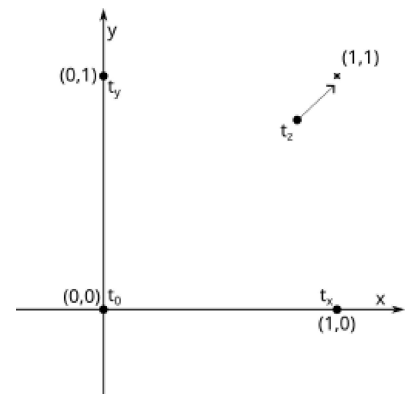
$$b_z = \mathbf{L}_t \mathbf{T}_t(t_z)$$

$$X = b_{zx}$$

$$Y = b_{zy}$$

$$\mathbf{P}_t^{-1} = \left[\begin{array}{cc|c} X & 0 & 0 \\ 0 & Y & 0 \\ \hline 1-Y & 1-X & X+Y-1 \end{array} \right] \quad \mathbf{P}_t = \left[\begin{array}{cc|c} Y \cdot (X+Y-1) & 0 & 0 \\ 0 & X \cdot (X+Y-1) & 0 \\ \hline Y \cdot (Y-1) & X \cdot (X-1) & X \cdot Y \end{array} \right]$$

$$\mathbf{P}_t^{-1} : \begin{array}{|c|c|c|c|} \hline -() & + & - & \cdot \\ \hline 0+0+3 & 0+2+1 & 2+0+3 & 0+4+0 \\ \hline \end{array}$$



Homographies, Assemble

Final matrix can be calculated:

$$\mathbf{H} = (\mathbf{P}_t \mathbf{L}_t \mathbf{T}_t)^{-1} \mathbf{P}_s \mathbf{L}_s \mathbf{T}_s = \mathbf{T}_t^{-1} \mathbf{L}_t^{-1} \mathbf{P}_t^{-1} \mathbf{P}_s \mathbf{L}_s \mathbf{T}_s$$

[Heckbert99] solution by using presented elements:

$$\text{Case 1 : } \mathbf{C}_1 = \mathbf{T}_t^{-1} \circ \mathbf{L}_t^{-1} \circ \mathbf{P}_t^{-1}$$

$$\text{Case 2 : } \mathbf{C}_2 = \mathbf{P}_s \circ \mathbf{L}_s \circ \mathbf{T}_s$$

Total usage of numerical operations:

f	-()	+	-	·
$\mathbf{T}_s$	2	0	0	0
$\mathbf{L}_s$	0+2	0+0	4+1	0+2
$\mathbf{P}_s$	0+0+0	0+2+2	2+0+4	0+4+4
$\mathbf{T}_t^{-1}$	0+0+0	0+0+0	0+0+0	0+0+0
$\mathbf{L}_t^{-1}$	0+0+0	0+0+0	0+4+0	0+0+0
$\mathbf{P}_t^{-1}$	0+0+3	0+2+1	2+0+3	0+4+0
$\mathbf{H}$	$5 \cdot 0$	$5 \cdot 4$	$5 \cdot 0$	$5 \cdot (4 \cdot 2)$
Σ	7	27	20	54

Higher Dimensions

Equalities

$$(X,Y) = \mathbf{P_t^{-1}}(1,1);$$

$$(X,Y,Z) = \mathbf{P_t^{-1}}(1,1,1);$$

$$(X,Y,Z,T) = \mathbf{P_t^{-1}}(1,1,1,1);$$

$$(X,Y,Z,T,S) = \mathbf{P_t^{-1}}(1,1,1,1,1);$$

$$\mathbf{b_z} = \mathbf{P_t^{-1}}(1,...);$$

0

Nothing to see here

1

$$\mathbf{P_t^{-1}} = \left[\begin{array}{c|c} X & 0 \\ \hline X & 1-X \end{array} \right] \quad \mathbf{P_t} = \left[\begin{array}{c|c} (X-1) & 0 \\ \hline X \cdot (X-1) & X \end{array} \right]$$

2

$$\mathbf{P_t^{-1}} = \left[\begin{array}{cc|c} X & 0 & 0 \\ 0 & Y & 0 \\ \hline 1-Y & 1-X & X+Y-1 \end{array} \right] \quad \mathbf{P_t} = \left[\begin{array}{cc|c} Y \cdot (X+Y-1) & 0 & 0 \\ 0 & X \cdot (X+Y-1) & 0 \\ \hline Y \cdot (Y-1) & X \cdot (X-1) & X \cdot Y \end{array} \right]$$

3

$$\mathbf{P_t^{-1}} = \left[\begin{array}{ccc|c} 2 \cdot X & 0 & 0 & 0 \\ 0 & 2 \cdot Y & 0 & 0 \\ 0 & 0 & 2 \cdot Z & 0 \\ \hline 1+X-Y-Z & 1-X+Y-Z & 1-X+Y+Z & X+Y+Z-1 \end{array} \right]$$

$$\mathbf{P_t} = \left[\begin{array}{ccc|c} Y \cdot Z \cdot (X+Y+Z-1) & 0 & 0 & 0 \\ 0 & X \cdot Z \cdot (X+Y+Z-1) & 0 & 0 \\ 0 & 0 & X \cdot Y \cdot (X+Y+Z-1) & 0 \\ \hline Y \cdot Z \cdot (X+Y+Z-1-2 \cdot X) & X \cdot Z \cdot (X+Y+Z-1-2 \cdot Y) & X \cdot Y \cdot (X+Y+Z-1-2 \cdot Z) & 2 \cdot X \cdot Y \cdot Z \end{array} \right]$$

4

$$\mathbf{P_t^{-1}} = \left[\begin{array}{cccc|c} 3 \cdot X & 0 & 0 & 0 & 0 \\ 0 & 3 \cdot Y & 0 & 0 & 0 \\ 0 & 0 & 3 \cdot Z & 0 & 0 \\ 0 & 0 & 0 & 3 \cdot T & 0 \\ \hline 1+X-Y-Z-T & 1-X+Y-Z-T & 1-X+Y+Z-T & 1-X-Y-Z+T & X+Y+Z+T-1 \end{array} \right]$$

5

$$\mathbf{P_t^{-1}} = \left[\begin{array}{ccccc|c} 4 \cdot X & 0 & 0 & 0 & 0 & 0 \\ 0 & 4 \cdot Y & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 \cdot Z & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \cdot T & 0 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 4 \cdot S & 0 \\ \hline 1+X-Y-Z-T-S & 1-X+Y-Z-T-S & 1-X-Y+Z-T-S & 1-X-Y-Z+T-S & 1-X-Y-Z-T+S & X+Y+Z+T+S-1 \end{array} \right]$$

n

$$\mathbf{P_t^{-1}} = \left[\begin{array}{ccccc|c} (n-1) \cdot b_{z_1} & 0 & \dots & \dots & 0 & 0 \\ 0 & \dots & 0 & \dots & \dots & 0 \\ \dots & 0 & (n-1) \cdot b_{z_i} & 0 & \dots & 0 \\ \dots & \dots & 0 & \dots & 0 & 0 \\ 0 & \dots & \dots & 0 & (n-1) \cdot b_{z_n} & 0 \\ \hline \dots & \dots & 1+b_{z_i} - \sum_{j=1, j \neq i}^n b_{z_j} & \dots & \dots & (-1) + \sum_{j=1}^n b_{z_i} \end{array} \right]$$

$$0 \neq (-1) + b_{z_i} - \sum_{j=1}^n b_{z_j}$$

$$\mathbf{P_t} = \left[\begin{array}{ccccc|c} \dots & 0 & \dots & \dots & 0 & 0 \\ 0 & \dots & 0 & \dots & \dots & 0 \\ \dots & 0 & \left(\prod_{j=1, j \neq i}^n b_{z_j} \right) \left((-1) + \sum_{j=1}^n b_{z_j} \right) & 0 & \dots & 0 \\ \dots & \dots & 0 & \dots & 0 & 0 \\ 0 & \dots & \dots & 0 & \dots & 0 \\ \hline \dots & \dots & \left(\prod_{j=1, j \neq i}^n b_{z_j} \right) \left((n-1) \cdot b_{z_i} + \sum_{j=1}^n b_{z_j} \right) & \dots & \dots & (n-1) \prod_{j=1}^n b_{z_i} \end{array} \right]$$

$$0 \neq \prod_{j=1}^n b_{z_i}$$

Constraints

If we use any tree point and try to make affine transform those tree points must make triangle with non zero area.

For higher dimension constraints are similar. Any volume must be non zero.

3d:

- $volume(s_i, s_j, s_k, s_l) \neq 0; i \neq j \neq k \neq l; i, j, k, l \in \{0, 1, 2, 3, 4\}$
- $volume(t_i, t_j, t_k, t_l) \neq 0; i \neq j \neq k \neq l; i, j, k, l \in \{0, 1, 2, 3, 4\}$

Additional requirements previously mentioned are sub set of this one.

Shortcut

Instead to have: $t_z = \mathbf{P_t}^{-1} (\mathbf{P_t}(s_z))$, why not use: $t_z = (\mathbf{P_t}^{-1} \circ \mathbf{P_t})(s_z)$

Appears that formula for that still can be written in readable form.

2

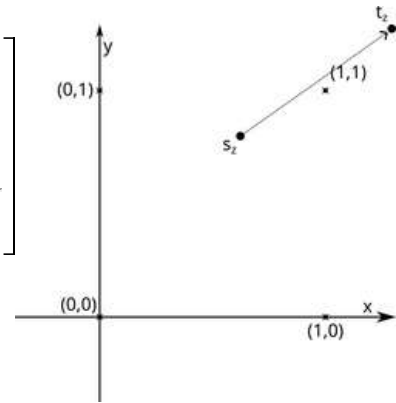
$$\mathbf{P_t}^{-1} \circ \mathbf{P_t} = \left[\begin{array}{cc|c} t_x \cdot (s_x + s_y - 1) \cdot s_y & 0 & 0 \\ 0 & t_y \cdot (s_x + s_y - 1) \cdot s_x & 0 \\ \hline (t_x \cdot (s_y - 1) - s_x \cdot (t_y - 1)) \cdot s_y & (t_y \cdot (s_x - 1) - s_y \cdot (t_x - 1)) \cdot s_x & (s_x \cdot s_y) \cdot (t_x + t_y - 1) \end{array} \right]$$

n-dimensional matrix is:

$$\mathbf{P_t}^{-1} \circ \mathbf{P_t} = \left[\begin{array}{ccc|c} \dots & 0 & \dots & 0 \\ \dots & \dots & t_i \cdot \sum_s \cdot \prod_{j \neq i} s_j & \dots \\ 0 & 0 & \dots & 0 \\ \hline \dots & \dots & (t_i \cdot (\sum_s - s_i) - s_i \cdot (\sum_t - t_i)) \cdot \prod_{j \neq i} s_j & \dots \end{array} \right] \left| \begin{array}{c} 0 \\ 0 \\ 0 \\ \prod_j s_j \cdot \sum_t \end{array} \right.$$

$$\sum_s = (-1) + \sum s_i$$

$$\sum_t = (-1) + \sum t_i$$



Miscellaneous

***this**

Latest version of this document can be found at:

<https://github.com/dmilos/math/blob/master/doc/homography/nhd.html>

Source code

c++

<https://github.com/dmilos/math/blob/master/src/math/linear/homography/construct1.hpp>
<https://github.com/dmilos/math/blob/master/src/math/linear/homography/construct2.hpp>
<https://github.com/dmilos/math/blob/master/src/math/linear/homography/construct3.hpp>
<https://github.com/dmilos/math/blob/master/src/math/linear/homography/constructn.hpp>

Maxima

<https://github.com/dmilos/math/blob/master/doc/homography/construct.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-2d.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-2d-zero-s.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-2d-zero-t.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-3d.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-3d-check.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-3d-zero-s.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-3d-zero-t.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-4d.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-4d-zero-t.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-5d.max>
<https://github.com/dmilos/math/blob/master/doc/homography/construct-practical.max>

Off topic

Homography matrix can be multiplied by some non zero factor for better utilization.

Here are several proposed values to do that:

$$\lambda = 1/h_{33}$$

This is one of the most common way to do. Numerical error may occur if h_{33} is very small comparing to h_{31} and h_{32}

$$\lambda = s / \|h_3\|, s \in \mathbb{R} \ s \neq 0;$$

Another common way to do. s is usually equal to 1.

$$\lambda = s / \|h_{*1}\|, s \in \mathbb{R} \ s \neq 0;$$

Observe first column as 3D vector. s is usually equal to 1. First column can be seen as first basis vector of first plane.

$$\lambda = 1 / \|h_{*1} \times h_{*2}\|$$

$h_{*1} \times h_{*2}$ is normal of source plane.

$$\lambda = 1 / \|v_x \cdot v_x + v_y \cdot v_y\|; v = h_{*1} \times h_{*2};$$

This cross product also give equation of horizon or vanish line on target plane($z=1$).

First and second coordinate will present direction of line in form: $x \cdot \cos(\alpha) + y \cdot \sin(\alpha) - r = 0$

$$\lambda = h_{*3} \cdot (h_{*1} \times h_{*2})$$

Distance of plane from origin.

Links

- <https://en.wikipedia.org/wiki/Homography>
- [Heckbert89] *Fundamentals of Texture Mapping and Image Warping*, Paul S. Heckbert, 17 June 1989.
- [Heckbert99] *Projective Mappings for Image Warping*, Paul Heckbert, 15-869, Image-Based Modeling and Rendering, 13 Sept 1999.